Genetic Lifehacks
Learn. Experiment. Optimize.

COMT Report

COMT

Catechol-O-methyltransferase

Gene Function:
The catechol-O-methyltransferase (COMT) gene plays a critical role in the metabolic breakdown of catechols, including neurotransmitters dopamine, epinephrine, and norepinephrine, estrogen metabolites, and other substances with a catechol structure. Thus, COMT helps to control neurotransmitter levels and protect cells from oxidative stress.

Methyl-Donor Supplements and COMT:
The reactions using the COMT enzyme involve adding a methyl group to a substance to synthesize a new substance. Increasing methyl groups via high-dose methyl donor supplements can enhance COMT reactions. However, adding a lot of methyl donors may affect mood in people with slow COMT. Methyl donor supplements include SAMe (s-adenosylmethionine), Methylfolate, Methylcobalamin (methylB12), and TMG/betaine.

COMT and Supplement Interactions:
Supplements that inhibit COMT function include natural flavonoids such as quercetin, fisetin, luteolin, rutin, and oleacein. COMT is also used in the metabolism of EGCG, although its inhibitory effect on COMT in humans is less clear. Such interactions typically require high levels found in supplements.

SNP ID	EFFECT ALLELE	YOUR GENOTYPE	STATUS
rs4680	A	**AG**	Heterozygous

Genetic Lifehacks Personalized Report

Understanding your data:

Intermediate COMT Function:

Your genotype shows that you have one copy of the faster variant and one copy of the slower variant. This means that you likely have an intermediate function for COMT. This is the most common genotype, in most population groups.Individuals with intermediate COMT function usually are able to take methyl donor supplements in reasonable doses. If you find that methyl donor supplements cause irritability or anxiety, consider lowering the dose or looking for alternatives. Alternatives to methylB12 include hydroxyB12 or adenosylB12. Alternatives to methylfolate include adding more folate to your diet, increasing dietary choline, supplementing with folinic acid (low dose), or supplementing with creatine.

References: Please see the complete COMT article here for a full list of references and a more detailed explanation: https://www.geneticlifehacks.com/comt-genetic-connections-to-neurotransmitter-levels/

Disclaimer: This report is for informational and educational purposes only. It is not intended to treat, diagnose, or cure any condition. Please talk with your healthcare provider if you have any questions on supplements or vitamins.